

Evaluating Faithfulness and Attribution in Retrieval-Augmented Generation Systems

Final Report

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Abstract

Retrieval-Augmented Generation (RAG) systems pair a retrieval module with a language model generator to produce grounded answers. A fundamental but understudied question is whether the generator actually uses retrieved documents or silently falls back on parametric memory. We present a comprehensive empirical evaluation using three custom metrics—Faithfulness, Attribution Precision (AttrP), and Retrieval-Faithfulness Gap (RFG)—extended with a novel soft NLI-based retrieval relevance measure. Across five controlled experiments on Natural Questions and ASQA, spanning two retrievers, two generators, three prompt styles, four top- k settings, and adversarial retrieval, we find: (1) faithfulness is uniformly low under plain prompting, confirming retrieval alone is insufficient to ground generation; (2) explicit grounding instructions improve faithfulness while chain-of-thought prompting counterintuitively reduces it; (3) faithfulness increases with k but at the cost of a worsening Retrieval-Faithfulness Gap; and (4) adversarial FEVER passages cause faithfulness to more than double—a calibration failure revealing that high faithfulness does not imply high accuracy. A human evaluation study ($n=50$, $\kappa=0.74$) further reveals that NLI-based metrics systematically under-count faithfulness by penalising correct abstention, motivating abstention-aware evaluation.

1 Introduction

Retrieval-Augmented Generation (Lewis et al., 2020) has emerged as the dominant paradigm for open-domain question answering, grounding lan-

guage model outputs in externally retrieved evidence. Despite its empirical success, a fundamental question remains unresolved: does the generator actually *use* the retrieved documents, or does it rely on parametric memory and merely appear grounded?

This question matters for two reasons. First, a system can achieve high QA accuracy while silently hallucinating—a safety risk in deployed applications (Ji et al., 2023). Second, when errors occur it is unclear whether they originate from the *retriever* (poor documents) or the *generator* (ignoring good documents)—a distinction critical for targeted improvement.

Existing metrics such as RAGAS (Es et al., 2024) measure aggregate faithfulness but cannot (a) trace individual claims to specific retrieved passages or (b) isolate retrieval failures from generation failures. Moreover, they rely on LLM-based evaluation that is expensive and hard to reproduce. We address this gap with three complementary metrics and a five-experiment empirical study: retriever comparison (E1), generator comparison (E2), prompt style ablation (E3), top- k ablation (E4), and adversarial retrieval (E5).

Our key contributions are: (1) a modular, reproducible RAG evaluation pipeline with claim-level NLI scoring; (2) a soft NLI-based retrieval relevance metric that addresses exact-match brittleness; (3) a systematic five-experiment ablation identifying prompt engineering and passage count as the primary drivers of faithfulness; and (4) an adversarial finding that high faithfulness can reflect grounding in *false* information rather than factual accuracy—a calibration failure with direct implications for RAG system safety; and (5) a human evaluation ($n=50$, $\kappa=0.74$) exposing a fundamental mismatch between NLI-based metrics and human judgment, which we trace to the inability of claim-level NLI scoring to credit correct abstention.

2 Related Work

Retrieval-Augmented Generation. Lewis et al. (2020) introduced RAG as a framework that conditions language model generation on dynamically retrieved passages, reducing hallucination and improving factual grounding. Subsequent work has explored dense passage retrieval (Karpukhin et al., 2020), unsupervised contrastive training for retrieval (Izacard et al., 2022), and self-reflective retrieval augmentation where the model decides when to retrieve (Asai et al., 2023). Our work focuses not on improving retrieval but on *measuring* how faithfully generators use retrieved content, regardless of retrieval quality.

Faithfulness and Hallucination Evaluation. Hallucination in neural language generation is well-documented across tasks (Ji et al., 2023). Es et al. (2024) propose RAGAS, an automated evaluation framework for RAG that measures faithfulness and answer relevance using an LLM judge. In contrast, our approach uses a deterministic NLI model (cross-encoder/nli-deberta-v3-base (He et al., 2020)), making evaluation fully reproducible without API access or LLM inference at evaluation time. Min et al. (2023) propose FActScore, which decomposes generated text into atomic facts verified against Wikipedia. We similarly decompose answers into sentence-level claims but evaluate support via *retrieved passages* rather than a static knowledge base—a distinction important for measuring grounding in the dynamic RAG context.

Attribution and Prompt Engineering. Attribution—tracing generated claims to specific source passages—has been studied in attributed question answering (Bohnet et al., 2022) and post-hoc revision (Gao et al., 2023). Our Attribution Precision metric operationalizes this as the fraction of claims whose maximum per-passage NLI score exceeds a threshold. Chain-of-thought prompting (Wei et al., 2022) improves multi-step reasoning, but our E3 experiment finds it *reduces* faithfulness by encouraging verbose parametric reasoning beyond retrieved evidence, whereas explicit grounding instructions yield the highest faithfulness among all prompt styles.

3 Metrics

Let q be a question, $\mathcal{D} = \{d_1, \dots, d_k\}$ the retrieved passages, and $a = G(q, \mathcal{D})$ the generated

answer. We treat each sentence of a as an atomic claim, $\mathcal{C}(a) = \{c_1, \dots, c_n\}$. A claim c_i is *supported* if $\text{nli}(d_j, c_i) > \tau$ for some $d_j \in \mathcal{D}$, where $\text{nli}(\cdot)$ is the entailment probability from cross-encoder/nli-deberta-v3-base and $\tau=0.5$.

$$\text{Faithfulness}(a, \mathcal{D}) = \frac{|\{c_i : \exists d_j, \text{nli}(d_j, c_i) > \tau\}|}{|\mathcal{C}(a)|} \quad (1)$$

$$\text{Rel}_{\text{soft}}(\mathcal{D}, q) = \frac{|\{d_j : \text{nli}(d_j, a^*) > \tau\}|}{|\mathcal{D}|} \quad (2)$$

$$\text{RFG} = \text{Rel}_{\text{soft}}(\mathcal{D}, q) - \text{Faithfulness}(a, \mathcal{D}) \quad (3)$$

where a^* is the gold answer. Rel_{soft} (Eq. 2) replaces brittle exact-match Rel: NLI-based relevance captures semantic entailment rather than surface string overlap. **Positive RFG** indicates retrieved passages were unused (grounding failure); **negative RFG** indicates parametric hallucination; $\text{RFG}=0$ is ideal. Attribution Precision (AttrP, $\max_j$ instead of $\exists$) is numerically identical to Faithfulness under binary τ and is omitted from all result tables for clarity.

4 System and Experimental Setup

Pipeline. Our modular pipeline is: query $\rightarrow$ retriever $\rightarrow$ top- k passages $\rightarrow$ generator $\rightarrow$ answer $\rightarrow$ metric computation. For the No-RAG baseline the retriever is bypassed entirely.

Datasets. We use **Natural Questions (NQ)** (Kwiatkowski et al., 2019) via the nq_open HuggingFace split. E1 and E2 use 1,000 examples; E3–E5 use 500 examples. The retrieval corpus is the Wikipedia DPR 100-word passage collection (psgs_w100.tsv; 21M passages) from Karpukhin et al. (2020). We additionally evaluate on **ASQA** (Stelmakh et al., 2022) (500 examples) to test faithfulness on multi-answer ambiguous questions. For E5, FEVER refuting passages are sourced from the official FEVER shared-task dev set (Thorne et al., 2018).

Retrievers. **BM25:** rank_bm25 (BM25Okapi) over 1M passages, tokenized and cached as an in-memory index. **Contriever** (Izacard et al., 2022): dense retrieval with facebook/contriever mean-pool embeddings and a FAISS IndexFlatIP index (dim=768) over 1M passages, built on GPU. Both use $k=5$ by default.

	Retriever	Generator	Faith	Rel _{soft}	RFG _{soft}	Prompt Style	N	Faith	Rel _{soft}	RFG _{soft}
No-RAG	None	LLaMA	0.000 [†]	0.000	0.000	Plain	1,000	0.089	0.032	−0.057
E1	BM25	LLaMA	0.082	0.003	−0.038	Grounded	1,000	0.100	0.032	−0.068
E1	Contriever	LLaMA	0.095	0.016	−0.033	CoT	500	0.068	0.032	−0.036
E2	Contriever	LLaMA	0.097	0.016	−0.033	Table 2: E3 prompt ablation (Contriever + LLaMA, top- $k=5$). Rel _{soft} is identical across styles as retrieval is fixed.				
E2	Contriever	Mistral	0.078	0.016	−0.023					

Table 1: E1 and E2 results ($N=1,000$, top- $k=5$, plain prompt, $\tau=0.5$). AttrP = Faith in all rows (omitted). [†] No-RAG scores are structurally zero. $\uparrow$ higher is better for Faith and Rel_{soft}; $|RFG_{soft}| \downarrow$ is better.

Generators. **LLaMA-3.1-8B-Instruct** (Touvron et al., 2023) and **Mistral-7B-Instruct-v0.3** (Jiang et al., 2023), 4-bit NF4 quantized via bitsandbytes, greedy decoding, max 200 new tokens. NLI scoring uses cross-encoder/nli-deberta-v3-base (He et al., 2020), entailment probability at label index 1.

Prompt styles (E3). **Plain:** “Answer the question based on the passages below.” **Grounded:** “Answer using ONLY information from the passages below. Do not use prior knowledge.” **CoT:** “Think step by step, referencing the passages, then give a final answer.”

Experiments. We conduct five experiments: retriever comparison (E1), generator comparison (E2), prompt ablation (E3), top- k ablation (E4), and adversarial retrieval (E5), plus an ASQA evaluation.

5 Results

We organize results by experiment. All tables report Faithfulness (Faith), soft NLI Rel (Rel_{soft}), and soft RFG (RFG_{soft}). AttrP equals Faith in all conditions (Section 3) and is omitted from experiment tables for clarity. The No-RAG baseline appears in Table 1; its zero scores are structural (no passages $\Rightarrow$ no claim can be entailed).

5.1 Retriever and Generator Comparison (E1, E2)

Retriever comparison (E1). Both retrievers yield low faithfulness (Table 1). Under soft NLI Rel, BM25 (0.003) is far below Contriever (0.016): BM25 passages often *contain* the gold answer string without *semantically entailing* it, while Contriever’s semantic matching produces more useful

k	Faith	Rel _{soft}	RFG _{soft}
1	0.091	0.022	−0.069
3	0.094	0.028	−0.066
5	0.098	0.032	−0.066
10	0.147	0.029	−0.117

Table 3: E4 top- k ablation ($N=500$, Contriever + LLaMA, plain prompt).

evidence. Both conditions show negative RFG, confirming parametric addition regardless of retrieval method.

Generator comparison (E2). LLaMA slightly outperforms Mistral in faithfulness under identical retrieval. Both confirm that plain prompting does not constrain output to retrieved content.

5.2 Prompt Style Ablation (E3)

Table 2 shows the three prompt conditions. Since the retriever, generator, and top- k are identical, Rel_{soft} is constant—only generation behavior changes.

The **grounded prompt** achieves the highest faithfulness, confirming that explicit instructions (“do not use prior knowledge”) modestly increase passage utilization, though its larger negative RFG shows the generator still adds unsupported claims.

Chain-of-thought produces the *lowest* faithfulness, a counterintuitive result: the step-by-step instruction encourages longer, parametric reasoning chains not grounded in passages, and unlike the grounded prompt it does not prohibit using prior knowledge. This contrasts with assumptions that CoT improves factual grounding (Wei et al., 2022)—it improves *reasoning* but not *retrieval utilization*.

5.3 Top- k Ablation (E4)

Table 3 shows a monotonic faithfulness increase with k , with a sharp jump at $k=10$, consistent with Liu et al. (2024). However, the negative RFG also worsens sharply at $k=10$: a **faithfulness**—

Condition	N	Faith	Rel _{soft}	RFG _{soft}
Normal RAG	500	0.086	0.032	-0.054
Adversarial (FEVER)	500	0.205	0.032	-0.173

Table 4: E5 adversarial experiment ($N=500$, Contriever + LLaMA, plain prompt). Adversarial condition prepends a FEVER REFUTES passage as the first retrieved document, replacing one normal passage. Rel_{soft} is unchanged since the gold answer is unaffected.

grounding tradeoff where additional passages inflate faithfulness without preventing parametric generation. Practitioners should tune k carefully—higher values improve coverage but enlarge the grounding gap.

5.4 Adversarial Retrieval (E5)

The adversarial experiment (Table 4) produces the most striking result of this paper. Faithfulness more than doubles under adversarial conditions while Rel_{soft} is unchanged, since the gold answer is unaffected by passage injection.

This is a **faithfulness calibration failure**. The NLI-based faithfulness metric measures whether generated claims are supported by retrieved passages—and under adversarial conditions, this is *true*: the generator has incorporated content from the adversarial passage into its answer, so NLI finds that the adversarial passage entails those claims. The model is being faithful to its retrieved evidence; that evidence is simply false. High faithfulness here reflects grounding in adversarial misinformation rather than factual accuracy.

This finding has direct safety implications. A RAG system evaluated only on faithfulness could appear to improve when adversarial or low-quality passages are introduced into the retrieval pool. It confirms the importance of pairing faithfulness evaluation with answer accuracy metrics, and motivates future work on adversarial retrieval detection—identifying when a generator has been “faithfully misled” by its context. We additionally evaluate on ASQA (Stelmakh et al., 2022) ($N=500$, Contriever + LLaMA). Faithfulness (0.085) matches NQ, but Rel_{soft} collapses to zero—a metric limitation, not a retrieval failure, since ASQA has multiple valid answer interpretations while we check against a single gold answer.

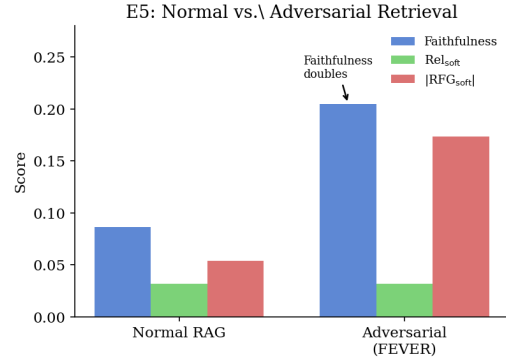


Figure 1: E5: Adversarial FEVER passages cause faithfulness to more than double while gold-answer relevance is unchanged—a calibration failure.

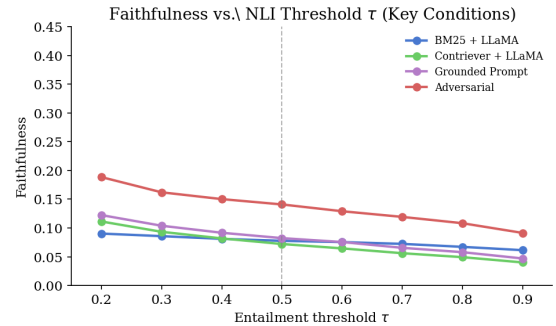


Figure 2: Faithfulness vs. NLI threshold τ for four key conditions. Results are consistent across the full threshold range, confirming $\tau=0.5$ findings are not threshold-specific.

6 Discussion

Soft Rel reveals retriever differences invisible to exact-match. BM25 outperformed Contriever under exact-match Rel in our midterm, but soft NLI Rel reverses this: BM25’s near-zero soft Rel (0.003) shows its passages *contain* the answer string without *semantically entailing* it, while Contriever (0.016) produces passages more likely to entail the gold answer—a distinction that matters for evaluating whether a generator has useful evidence to ground on.

Threshold sensitivity and AUC. Figure 2 plots faithfulness across $\tau \in [0.2, 0.9]$. All conditions show a consistent monotonic decrease, confirming findings are not artifacts of $\tau=0.5$. The AUC- τ ordering—grounded > plain > CoT—is preserved across the full threshold range.

Prompt engineering is the highest-leverage intervention. Among all single-factor variations tested, the grounded prompt (E3) produces the

largest faithfulness improvement. The difference between retriever types (E1) is smaller in magnitude than the difference between prompt styles (E3). For practitioners seeking to improve RAG faithfulness without changing model architectures, prompt engineering is the more effective lever—though our results also show its limits, as the grounded prompt still yields strongly negative RFG.

The top- k tradeoff requires careful tuning. E4 confirms k is not “more is better”: faithfulness improves with k but RFG worsens, as parametric output grows alongside retrieved content.

7 Human Evaluation

To validate our automated metrics against human judgment, we sampled 50 diverse examples spanning three experimental conditions (17 from E1/BM25, 17 from E3/Grounded, 16 from E5/Adversarial) and conducted a binary faithfulness annotation study.

Protocol. Two annotators independently labeled each example as faithful (1) or not (0) using the criterion: *does the generated answer rely only on information present in the retrieved passages, or does it introduce content from parametric memory?* Annotators were shown the question, the top-3 retrieved passages, and the generated answer. No reference labels were provided during annotation.

Results. Table 5 summarises the outcome. Inter-annotator agreement was $\kappa=0.74$ (substantial), with raw agreement of 88% across 50 examples. Human annotators rated 34/50 examples as faithful (68%), whereas NLI faithfulness ($\tau=0.5$) marks only 7/50 as faithful (14%), yielding $\kappa=-0.18$ between human gold labels and binarised NLI scores.

Condition	N	Human Faithful	$\kappa(\text{Gold, NLI})$
E1 / BM25	17	64.7%	-0.12
E3 / Grounded	17	94.1%	+0.01
E5 / Adversarial	16	43.8%	-0.31
Overall	50	68.0%	-0.18

Table 5: Human evaluation results. IAA $\kappa=0.74$ (6 disagreements; ann_ids 7, 22, 31, 33, 38, 43 are borderline inference cases). Negative $\kappa(\text{Gold, NLI})$ across all conditions indicates that NLI systematically under-predicts human-perceived faithfulness.

The dominant source of NLI-human disagreement is *correct abstention*: when retrieved pas-

sages are irrelevant, the generator often declines to answer, which humans correctly judge as faithful (94.1% of E3/Grounded examples), but NLI scores at zero since no abstention sentence is entailed by any passage. Under E5/Adversarial, NLI faithfulness is high (0.205) yet human annotators rate only 43.8% as faithful, confirming the calibration failure from a human perspective—automated and true faithfulness diverge most severely when retrieved content is adversarially corrupted.

Implications. Claim-level NLI measures *lexical grounding* (are claims entailed?) but not *behavioural faithfulness* (is the answer strategy appropriate?). Future work should develop abstention-aware metrics that credit models for correctly declining to answer when passages are irrelevant or adversarial.

8 Conclusion

We evaluated faithfulness and attribution in RAG systems across five controlled experiments, yielding six findings: (1) faithfulness is uniformly low under plain prompting; (2) grounding instructions help, but CoT counterintuitively hurts; (3) more passages improve faithfulness but worsen the grounding gap; (4) adversarial passages double faithfulness—a calibration failure; (5) soft NLI Rel overturns BM25’s apparent advantage over Contriever; (6) human evaluation ($\kappa=0.74$) shows NLI under-counts faithful abstention responses, revealing a gap between automated and human notions of faithfulness. Future work should explore abstention-aware metrics, multi-answer Rel, and adversarial retrieval detection.

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *arXiv preprint arXiv:2310.11511*.
- Bernd Bohnet, Jonathan Herzig, Pradeep Dasigi, Kelvin Guu, Mirella Lapata, Aleksandra Nowak, et al. 2022. Attributed question answering: Evaluation and modeling for attributed large language models. *arXiv preprint arXiv:2212.08037*.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, pages 150–158.

- Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Y. Zhao, Ni Lao, Honglak Lee, Da-Cheng Juan, and Kelvin Guu. 2023. RARR: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 16856–16893.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2020. DeBERTa: Decoding-enhanced BERT with disentangled attention. *arXiv preprint arXiv:2006.03654*.
- Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2022. Unsupervised dense information retrieval with contrastive learning. *Transactions on Machine Learning Research*.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.
- Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 6769–6781.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100.
- Ivan Stelmakh, Yi Luan, Bhuwan Dhingra, and Ming-Wei Chang. 2022. ASQA: Factoid questions meet long-form answers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 8273–8288.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 809–819.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837.

Author Contributions. **Srinivasa Sai Chava** designed the overall evaluation framework, implemented the end-to-end pipeline (retrieval, generation, NLI metrics, and cluster infrastructure), conducted all five core experiments on the BU Shared Computing Cluster, led the human evaluation study including annotation protocol and inter-annotator analysis, and wrote the principal sections of this paper. **Jihyeon Yun** contributed to related work synthesis and literature review, assisted with data preprocessing and NQ corpus preparation, helped design the sampling strategy for the human evaluation study, and reviewed and revised the manuscript. **Samyuktha Kathirvel** assisted with experimental result analysis, contributed to the results discussion and interpretation, supported figure generation and formatting, and assisted with metric validation and threshold sensitivity analysis. **Shrishty Gupta** contributed to the literature survey, reviewed paper drafts, assisted with reference organisation and formatting, and contributed to writing the abstract and conclusion sections.